Zhou Fan

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EDUCATION

2018 Ph.D. Statistics, Stanford University

2011 M.A.St. Mathematics, with distinction, University of Cambridge

2010 S.M. Computer Science, Harvard University

2010 A.B. Mathematics, summa cum laude, Harvard University

EMPLOYMENT

2025 – present Associate Professor, Statistics and Data Science, Yale University

2018 – 2025 Assistant Professor, Statistics and Data Science, Yale University

2011 – 2013 Scientific Programmer, D. E. Shaw Research, New York

PUBLICATIONS AND PREPRINTS

1. Max Lovig, Tianhao Wang, Zhou Fan. "On universality of non-separable Approximate Message Passing algorithms." Available at 2506.23010.

2. Zhou Fan, Justin Ko, Bruno Loureiro, Yue M. Lu, Yandi Shen. "Dynamical mean-field analysis of adaptive Langevin diffusions: Replica-symmetric fixed point and empirical Bayes." Available at arXiv:2504.15558.

3. Zhou Fan, Justin Ko, Bruno Loureiro, Yue M. Lu, Yandi Shen. "Dynamical mean-field analysis of adaptive Langevin diffusions: Propagation-of-chaos and convergence of the linear response." Available at arXiv:2504.15556.

4. Zhou Fan, Leying Guan, Yandi Shen, Yihong Wu. "Gradient flows for empirical Bayes in high-dimensional linear models." Available at arXiv:2312.12708.

5. Michael Celentano, Zhou Fan, Licong Lin, Song Mei. "Mean-field variational inference with the TAP free energy: Geometric and statistical properties in linear models." Available at arXiv:2311.08442.

6. Zhou Fan, Iain M. Johnstone, Yi Sun. "Spiked covariances and principal components analysis in high-dimensional random effects models." Available at arXiv:1806.09529.

7. Kaylee Yang, Timothy L. H. Wee, Zhou Fan. "Asymptotic mutual information in quadratic estimation problems over compact groups." *Information and Inference: Journal of the IMA*, to appear.

8. Zhou Fan, Renyuan Ma. “Kronecker-product random matrices and a matrix least squares problem.” *Annals of Probability*, to appear.

9. Zhou Fan, Yufan Li, Subhabrata Sen. “TAP equations for orthogonally invariant spin glasses at high temperature.” *Annales de l’Institut Henri Poincare*, to appear.

10. Zhou Fan, Yihong Wu. “The replica-symmetric free energy for Ising spin glasses with orthogonally invariant couplings.” *Probability Theory and Related Fields*, to appear.

11. Xinyi Zhong, Tianhao Wang, Zhou Fan. “Approximate Message Passing for orthogonally invariant ensembles: Multivariate non-linearities and spectral initialization.” *Information and Inference: Journal of the IMA*, 13(3): iaae024, 2024.

12. Tianhao Wang, Xinyi Zhong, Zhou Fan. “Universality of Approximate Message Passing algorithms and tensor networks.” *Annals of Applied Probability,* 34(4): 3943-3994, 2024.

13. Zhichao Wang, Denny Wu, Zhou Fan. “Nonlinear spiked covariance matrices and signal propagation in deep neural networks.” *Proceedings of the 37th Conference on Learning Theory (COLT)*, PMLR 247: 4891-4957, 2024.

14. Yufan Li, Zhou Fan, Subhabrata Sen, Yihong Wu. “Random linear estimation with rotationally-invariant designs: Asymptotics at high temperature.” *IEEE Transactions on Information Theory*, 30(3): 2118-2153, 2024.

15. Zehao Dou, Zhou Fan, Harrison Zhou. “Rates of estimation for high-dimensional multi-reference alignment.” *Annals of Statistics*, 52(1): 261-284, 2024.

16. Zhou Fan, Roy R. Lederman, Yi Sun, Tianhao Wang, Sheng Xu. “Maximum likelihood for high-noise group orbit estimation and single-particle cryo-EM.” *Annals of Statistics*, 52(1): 52-77, 2024.

17. Ran Cui, Roy A Elzur, Masahiro Kanai, Jacob C Ulirsch, Omer Weissbrod, Mark J Daly, Benjamin M Neale, Zhou Fan, Hilary K Finucane. “Improving fine-mapping by modeling infinitesimal effects.” *Nature Genetics*, 56: 162-169, 2024.

18. Michael Celentano, Zhou Fan, Song Mei. “Local convexity of the TAP free energy and AMP convergence for Z_2-synchronization.” *Annals of Statistics*, 51(2): 519-546, 2023.

19. Zhou Fan, Cheng Mao, Jiaming Xu, Yihong Wu. “Spectral graph matching and regularized quadratic relaxations I: Algorithm and Gaussian analysis.” *Foundations of Computational Mathematics*, 23: 1511-1565, 2023.

20. Zhou Fan, Cheng Mao, Jiaming Xu, Yihong Wu. “Spectral graph matching and regularized quadratic relaxations II: Erdos-Renyi graphs and universality.” *Foundations of Computational Mathematics*, 23: 1567-1617, 2023.

21. Zhou Fan, Yi Sun, Tianhao Wang, Yihong Wu. “Likelihood landscape and maximum likelihood estimation for the discrete orbit recovery model.” *Communications on Pure and Applied Mathematics*, 76(6): 1208-1302, 2023.

22. Zhou Fan, Iain M. Johnstone. “Tracy-Widom at each edge of real covariance and MANOVA estimators.” *Annals of Applied Probability*, 32(4): 2967-3003, 2022.

23. Xinyi Zhong, Chang Su, Zhou Fan. “Empirical Bayes PCA in high dimensions.” *Journal of the Royal Statistical Society: Series B*, 84(3): 853-878, 2022.

24. Zhou Fan. "Approximate Message Passing algorithms for rotationally invariant matrices." *Annals of Statistics,* 50(1): 197-224, 2022.

25. Zhou Fan, Yi Sun, Zhichao Wang. "Principal components in linear mixed models with general bulk." *Annals of Statistics*, 49(3): 1489-1513, 2021.

26. Sheng Xu, Zhou Fan. "Iterative Alpha Expansion for estimating gradient-sparse signals from linear measurements." *Journal of the Royal Statistical Society: Series B*, 83(2): 271-292, 2021.

27. Zhou Fan, Song Mei, Andrea Montanari. "TAP free energy, spin glasses, and variational inference." *Annals of Probability*, 49(1): 1-45, 2021.

28. Zhou Fan, Zhichao Wang. "Spectra of the Conjugate Kernel and Neural Tangent Kernel for linear-width neural networks." *Advances in Neural Information Processing Systems (NeurIPS)*, 33: 7710-7721, 2020. (Oral presentation)

29. Sheng Xu, Zhou Fan, Sahand Negahban. "Tree-projected gradient descent for estimating gradient-sparse parameters on graphs." *Proceedings of the 33rd Conference on Learning Theory (COLT)*, PMLR 125: 3683-3708, 2020.

30. Zhou Fan, Cheng Mao, Yihong Wu, Jiaming Xu. "Spectral graph matching and regularized quadratic relaxations: Algorithm and theory." *Proceedings of the 37th International Conference on Machine Learning (ICML)*, PMLR 119: 2985-2995, 2020.

31. Ganlin Song, Zhou Fan, John Lafferty. "Surfing: Iterative optimization over incrementally trained deep networks." *Advances in Neural Information Processing Systems (NeurIPS)*, 32, 2019. (Spotlight presentation)

32. Zhou Fan, Iain M. Johnstone. "Eigenvalue distributions of variance components estimators in high-dimensional random effects models." *Annals of Statistics*, 47(5): 2855-2886, 2019.

33. Zhou Fan, Andrea Montanari. "The spectral norm of random inner-product kernel matrices." *Probability Theory and Related Fields*, 173: 27-85, 2019.

34. Zhou Fan, Leying Guan. "Approximate l_0-penalized estimation of piecewise constant signals on graphs." *Annals of Statistics*, 46(6B): 3217-3245, 2018.

35. Leying Guan, Zhou Fan, Robert Tibshirani. "Supervised learning via the 'hubNet' procedure." *Statistica Sinica*, 28: 1225-1243, 2018.

36. Zhou Fan, Lester Mackey. "Empirical Bayesian analysis of simultaneous changepoints in multiple data sequences." *Annals of Applied Statistics*, 11(4): 2200-2221, 2017.

37. Zhou Fan, Andrea Montanari. "How well do local algorithms solve semidefinite programs?" *Proceedings of the 49th Annual ACM SIGACT Symposium on the Theory of Computing (STOC)*, 604-614, 2017.

38. Zhou Fan, Ron O Dror, Thomas Mildorf, Stefano Piana, David E Shaw. "Identifying localized changes in large systems: Change-point detection for biomolecular simulations." *Proceedings of the National Academy of Sciences USA* 112(24): 1-6, 2015.

39. Dominico Aiello, Hansheng Diao, Zhou Fan, Daniel King, Jessica Lin, and Cesar E Silva. "Measurable time-restricted sensitivity." *Nonlinearity* 25(12): 3313-3325, 2012.

INVITED TALKS AND SEMINARS

02/2026 Workshop on High-dimensional learning dynamics, Banff
08/2025 Math and Machine Learning seminar, Max Planck Institute + UCLA
08/2025 Joint Statistical Meetings, Nashville
07/2025 Stanford University, statistics seminar
07/2025 ICML Workshop on High-dimensional learning dynamics, Vancouver
06/2025 APS Informs Applied Probability Conference, Atlanta
06/2025 Workshop on Random matrices and high-dimensional learning dynamics, CRM Montreal
05/2025 Online Random Matrix Theory seminar (virtual)
05/2025 Workshop on Machine learning and neural network theory, EPFL
04/2025 Yale University, applied math seminar
03/2025 Workshop on Efficient Bayesian inference, BIRS Alberta
03/2025 Columbia University, statistics seminar
12/2024 University of Waterloo, probability seminar
11/2024 National Academy of Sciences Kavli Frontiers of Science Symposium, Beijing
09/2024 UCLA, level set seminar
09/2024 Workshop on Statistical aspects of non-linear inverse problems, Cambridge
08/2024 IMS World Congress in Probability and Statistics, Bochum
07/2024 Workshop on The blessing of dimensionality, Storrs
06/2024 Workshop on Modeling randomness in neural network training, DIMACS
05/2024 Random Matrices and Applications, ICERM
02/2024 Harvard University, statistics seminar
02/2024 Georgia Institute of Technology, stochastics seminar
01/2024 National University of Singapore, statistics seminar
01/2024 Program on Mathematics of Data, IMS Singapore
09/2023 Institute of Systems Science, Chinese Academy of Sciences
07/2023 Stochastic Processes and Applications, Lisbon
07/2023 Workshop on Learning and inference from structured data, ICTP Trieste
06/2023 APS Informs Applied Probability Conference, Paris
06/2023 Random Matrix Theory and Applications, Shenzhen
06/2023 Workshop on High dimensional statistics and random matrices, Porquerolles
11/2022 Workshop on Learning in networks: Performance limits and algorithms, BIRS Oaxaca
11/2022 UC Davis, probability seminar
10/2022 Brown University, applied mathematics (pattern theory) seminar
10/2022 MIT, stochastics and statistics seminar
10/2022 Wharton UPenn, statistics seminar
09/2022 SIAM Conference on Mathematics of Data Science, San Diego
06/2022 IMS Annual Meeting, London
06/2022 Workshop on Random matrix theory and numerical linear algebra, Seattle
05/2022 ICMS Workshop on Structural breaks and shape constraints, Edinburgh
03/2022 Boston University, probability and statistics seminar
10/2021 Rutgers University, statistics seminar
10/2021 Workshop on Computational complexity of statistical inference, Simons Institute
10/2021 HKUST, statistics seminar
09/2021 ICSA Applied Statistics Symposium
09/2021 Workshop on Noncommutative algebra, probability, and analysis in action, Greifswald
08/2021 IMSI Workshop on eliciting structure in genomics data, Chicago
07/2021 IMS World Congress in Probability and Statistics (virtual)
06/2021 Youth in High Dimensions workshop, ICTP Trieste (virtual)
06/2021 University of Warwick, statistics seminar
05/2021 Harvard University, Probabilitas seminar
05/2021 Workshop on RMT and machine learning (virtual)
01/2021 UC Davis, statistics seminar
12/2020 Workshop on Covariance matrices in large dimensions (virtual)
12/2020 ETH Zurich, Young Data Science Researcher Seminar
09/2020 USC, probability and statistics seminar
08/2020 Joint Statistical Meetings (virtual)

05/2020 Random Matrices and Their Applications (virtual)
12/2019 ICSA International Conference, Hangzhou
10/2019 NYU Stern, statistics seminar
09/2019 Allerton Conference, Urbana
09/2019 Cornell University, statistics seminar
07/2019 Joint Statistical Meetings, Denver
05/2019 New England Statistical Symposium, Hartford
05/2019 Workshop on Random matrices and related topics, Korea IAS
03/2019 Workshop on New developments in free probability and applications, CRM Montreal
02/2019 MIT, stochastics and statistics seminar
01/2019 Workshop on Universality in random structures, ICTS Bangalore
10/2018 Workshop on Robust and high-dimensional statistics, Simons Institute
07/2018 IMS annual meeting, Vilnius
06/2018 Workshop on Matrix estimation meets statistical network analysis, Oberwolfach
02/2018 NYU, Math and Data seminar
02/2018 Harvard University, statistics seminar
02/2018 Wharton UPenn, statistics seminar
02/2018 Carnegie Mellon University, statistics seminar
01/2018 Yale University, statistics seminar
01/2018 Columbia University, statistics seminar
01/2018 Booth School of Business, econometrics and statistics seminar
01/2018 University of Chicago, statistics seminar
01/2018 Texas A&M University, statistics seminar
01/2018 Duke University, statistical science seminar
01/2018 Princeton University, ORFE colloquim
01/2018 University of Michigan Ann Arbor, statistics seminar
11/2017 Georgia Institute of Technology, job candidate talk
11/2017 University of California Irvine, probability seminar
11/2017 AMS western fall sectional meeting, Riverside
04/2017 SLAC National Accelerator Laboratory, AI-AT-SLAC/FEL-R&D seminar
04/2017 Columbia University, Columbia-Courant probability day
04/2017 Georgia Institute of Technology, stochastics seminar
01/2017 Workshop on Random matrix theory and applications, Harvard CMSA
07/2016 IMS World Congress in Probability and Statistics, Toronto
04/2016 Stanford University, probability seminar

GRANTS

NSF DMS-2142476, “CAREER: High-dimensional inference and applications to modern biology.” Sole PI, $400,000, 2022-2027.

NSF DMS-1916198, “Non-convex landscapes and high-dimensional latent variable models.” Sole PI, $182,654, 2019-2022.

HONORS AND AWARDS

Sloan Research Fellowship, 2024
Arthur Greer Memorial Prize, Yale University, 2023
NSF CAREER Award, 2022
Probability Dissertation Award, Stanford Department of Statistics, 2018
Hertz Graduate Fellowship, 2013-2018
NDSEG Graduate Fellowship, 2013-2016
Churchill Scholarship, 2010-2011
Siebel Scholarship, 2009-2010

TEACHING

S&DS 242/542: Theory of Statistics
- Spring 2025. Enrollment: 191
- Spring 2022. Enrollment: 234
- Spring 2020. Enrollment: 225

S&DS 410/610: Statistical Inference
- Fall 2022. Enrollment: 40
- Fall 2021. Enrollment: 37
- Fall 2020. Enrollment: 59
- Fall 2019. Enrollment: 34
- Fall 2018. Enrollment: 31

S&DS 602: High-Dimensional Probability and Applications
- Fall 2024. Enrollment 26

S&DS 615: Introduction to Random Matrix Theory and Applications
- Spring 2019. Enrollment: 27

S&DS 686: High-Dimensional Phenomena in Statistics and Learning
- Spring 2023. Enrollment: 30

DOCTORAL AND POSTDOCTORAL TRAINEES

Supervised postdoctoral scholars:
- Yandi Shen, 2023-2024
  - First position: Carnegie Mellon University (assistant professor)

Supervised Ph.D. students:
- Garrett Wen (co-advised with Theodor Misiakiewicz), Yale S&DS, 2024-present
- Jack Ma, Yale S&DS, 2023-present
- Kaylee Yang (co-advised with Andre Wibisono), Yale S&DS, 2023-present
- Max Lovig (co-advised with Ilias Zadik), Yale S&DS, 2023-present
- Tianhao Wang, Yale S&DS, 2019-2024
  - First position: UC San Diego (assistant professor)
- Xinyi Zhong, Yale S&DS, 2019-2023
  - First position: Two Sigma Investments
- Chang Su (co-advised with Hongyu Zhao), Yale Biostatistics, 2019-2023
  - First position: Emory University (assistant professor)
- Sheng Xu (co-advised with Sahand Negahban), Yale S&DS, 2018-2022
  - First position: Princeton University (postdoc)

UNIVERSITY SERVICE

Supervised undergraduate research students and senior theses:
- David Dong, Spring 2024-present
- Jorge Blanco, Spring 2024
  - First position: UC Berkeley Statistics, Ph.D.
- Matthew Li, Fall 2023
- Sarah Zhao, 2020-2022
  - First position: Stanford Statistics, Ph.D.

Max Ranis, Spring 2022
Brian Zhu, Fall 2021
First position: Columbia IEOR, Ph.D.
Maya Jeyendran, Spring 2020
First position: Harvard Law School, J.D.
Ryan Taggarse, Spring 2019

Ph.D. dissertation committee member:
Leqi Xu, Yale Biostatistics (current)
Timothy Wee, Yale S&DS, 2024
Soham Jana, Yale S&DS, 2022
Yiliang Zhang, Yale BIS, 2022
Geyu Zhou, Yale CBB, 2022
Jerome Yu, Yale CBB, 2022
Zifan Li, Yale S&DS, 2021
Ganlin Song, Yale S&DS, 2021

Ph.D. oral exam committee member:
Chris Xu, Yale S&DS (current)
Xinyang Hu, Yale S&DS (current)
Heejune Sheen, Yale S&DS (current)
Awni Altabaa, Yale S&DS (current)
Conor Sheehan, Yale S&DS (current)

S&DS Director of Graduate Studies, 2024-present
S&DS Ph.D. admissions committee, Spring 2020, 2023, 2025
S&DS curriculum committee, 2021-present
S&DS department seminar organizer, 2018-2023
S&DS faculty search committee, Spring 2022
S&DS undergraduate thesis committee, Spring 2021, 2023
Foundations of Data Science Institute postdoctoral search committee, Spring 2023
Grace Hopper College freshman/sophomore advisor, 2019-2023

PROFESSIONAL SERVICE

K-12 educational outreach coordinator, Connecticut Chapter of the American Statistical Association, 2018-present

Editorial board:
Annals of Statistics, 2025 - present

Conference organization:
Joint Statistical Meetings 2024 (invited session organizer)
International Symposium on Nonparametric Statistics 2024 (invited session organizer)
Conference on Learning Theory 2022, 2023, 2024, 2025 (program committee)
Joint Statistical Meetings 2021 (invited session chair)

Grant reviewer:
US National Science Foundation, 2024
European Research Council, 2024

Journal reviewer:
Annals of Applied Probability
Annals of Applied Statistics
Annals of Statistics
Applied and Computational Harmonic Analysis

Biometrika
Biostatistics
Communications in Mathematical Physics
Communications on Pure and Applied Mathematics
Electronic Journal of Statistics
ESAIM: Probability and Statistics
Foundations of Computational Mathematics
IEEE Transactions on Information Theory
IEEE Transactions on Vehicular Technology
Information and Inference
Journal of Econometrics
Journal of Machine Learning Research
Journal of Mathematical Physics
Journal of Multivariate Analysis
Journal of the Royal Statistical Society: Series B
Linear and Multilinear Algebra
Mathematical Programming
Nature Genetics
Physical Review E
Probability and Mathematical Physics
Proceedings of the National Academy of Sciences USA
Random Matrices: Theory and Applications
SIAM Journal on Mathematics of Data Science
Statistics and Probability Letters
Stochastic Processes and their Applications

Conference reviewer:
AISTATS 2019
COLT 2022, 2023, 2024, 2025
ICML 2019
ISIT 2022
NeurIPS 2019, 2020, 2021, 2022
SODA 2018, 2020
STOC 2022